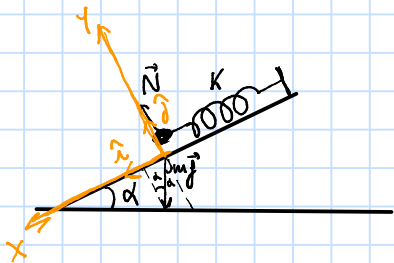


1)



$$a_x = g \sin \alpha - \omega^2 x \rightarrow \text{accelerazione tangenziale } a_t$$

$$-mg \cdot \hat{j} + N \cdot \hat{j} + \underbrace{F_{el} \cdot \hat{j}}_0 = m a \cdot \hat{j}$$

$$-mg \cos \alpha + N = 0 \rightarrow N = mg \cos \alpha$$

$$a_t \rightarrow \ddot{x} \rightarrow \begin{cases} \ddot{x} + \omega^2 x = g \sin \alpha \\ x(0) = 0 \\ \dot{x}(0) = 0 \end{cases}$$

$$\begin{cases} [x(t) - X_{\text{mnd}}] + \omega^2 [x(t) - X_{\text{mnd}}] = 0 \\ [x(t) - X_{\text{mnd}}]_{t=0} = -X_{\text{mnd}} \\ [x(t) - X_{\text{mnd}}]_{t=0} = 0 \end{cases}$$

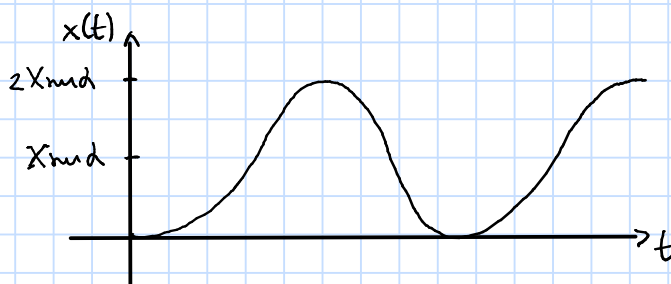
$$\ddot{x} + \omega^2 x - g \sin \alpha = 0 \rightarrow \ddot{x}(t) + \omega^2 \left[x(t) - \underbrace{\frac{g \sin \alpha}{\omega^2}}_{X_{\text{mnd}}} \right] = 0$$

$$\ddot{x}(t) + \omega^2 [x(t) - X_{\text{mnd}}] = 0 \quad \ddot{x}(t) = [x(t) - X_{\text{mnd}}]''$$

$$\omega^2 = \frac{k}{m}$$

$$X = \frac{mg}{k}$$

$$\rightarrow x(t) - X_{\text{mnd}} = -X_{\text{mnd}} \cos \omega t = X_{\text{mnd}} (1 - \cos \omega t)$$



2)



$$\vec{v}_0 = v_0 \hat{i}$$

$$\begin{cases} \ddot{x} + \omega^2 x = g \\ x(0) = 0 \\ \dot{x}(0) = v_0 \end{cases} \quad \begin{cases} [x(t) - X] + \omega^2 [x(t) - X] = 0 \\ [x(t) - X]_{t=0} = \cancel{0} - X \\ [\dot{x}(t) - \dot{X}]_{t=0} = v_0 \end{cases}$$

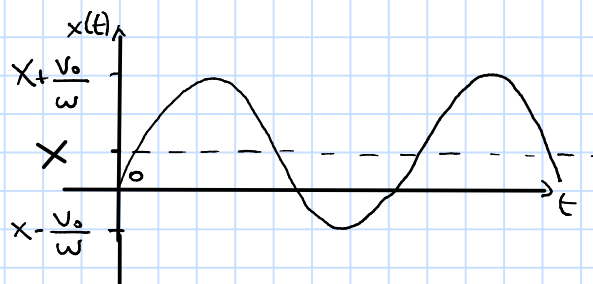
$$x(t) - X = \frac{v_0}{\omega} \sin \omega t \rightarrow X + \frac{v_0}{\omega} \sin \omega t$$

$$x(t) = X \left(1 + \frac{v_0}{\omega X} \sin \omega t \right)$$

$$-\frac{v_0}{\omega X} < \frac{v_0}{\omega X} \sin \omega t < \frac{v_0}{\omega X}$$

$$1 - \frac{v_0}{\omega X} < 1 + \frac{v_0}{\omega X} \sin \omega t < 1 + \frac{v_0}{\omega X}$$

$$X + \frac{v_0}{\omega} < x(t) < X - \frac{v_0}{\omega}$$



$$\omega t$$

$$T = \frac{2\pi}{\omega}$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\omega = \frac{2\pi}{T}$$

